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Euler-Cauchy $t^2y'' + \alpha t y' + \beta y = 0$ $t>0$ α, β constant

Seek soln $y = t^r$

$t^2 r(r-1)t^{r-2} + \alpha t r t^{r-1} + \beta t^r = 0$

$r^2 - r + \alpha r + \beta = 0$

Quadratic for r : $r^2 + r(\alpha - 1) + \beta = 0$

§3.8-1 $\sqrt{(\alpha-1)^2 - 4\beta} > 0 \longrightarrow r_1 \neq r_2$

$y_1 = t^{r_1}$ $y_2 = t^{r_2}$

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} t^{r_1} & t^{r_2} \\ r_1 t^{r_1-1} & r_2 t^{r_2-1} \end{vmatrix}$$

$= t^{r_1} t^{r_2-1} r_2 - t^{r_2} r_1 t^{r_1-1}$

$= t^{r_1+r_2-1}(r_2 - r_1)$

$\neq 0 \quad \forall t$

$(t^r \equiv e^{r \ln t} = e^{h t^r})$

Ex 3.25 $2t^2y'' + 3ty' - y = 0$ $t>0$

$y = t^r$ $y' = r t^{r-1}$ $y'' = r(r-1)t^{r-2}$

LHS : $2t^r r(r-1)t^{r-2} + 3t r t^{r-1} - t^r$

$t^r[2r^2 - 2r + 3r - 1] = 0$ RHS

if $(2r-1)(r+1) = 0$

$r_1 = 1/2$ $r_2 = -1$

general soln $y = c_1 t^{1/2} + c_2 t^{-1}$ $t>0$ //

§3.8-2 $y_1 = t^r$

Find y_2 use $R \neq 0$: $y = y_1 v(t) = y_1(t) v(t)$

$y' = y_1 v' + v y_1'$

$y'' =$

Sub into ODE to find an eq^n for $v(t)$. (3.45)

write $v' = W$ $v'' = W'$

(3.45) $t^2 W' = -t(2r+\alpha) W$ $W' = \frac{dW}{dt}$

$\int \frac{dW}{W} = -\int \frac{t(2r+\alpha)}{t^2} dt$ for $W \neq 0$

$$\ln|w| = -(2r+\alpha)\int t^{-1}\,dt$$

$$= -(2r+\alpha)\ln|t| + \ln c \qquad \text{constant}$$

$$\ln|v'| = -(2r+\alpha)\ln|t| + \ln c$$

$$e^{\ln|v'|} = |v'| = e^{-(2r+\alpha)\ln|t| + \ln c}$$

$$= e^{-(2r+\alpha)\ln|t|} \cdot e^{\ln c}$$

$$= e^{\ln|t|^{-(2r+\alpha)}} \cdot c$$

$$= |t|^{-(2r+\alpha)} \cdot c$$

$$= t^{-(2r+\alpha)} \cdot c \qquad \text{for } t>0$$

$$\Rightarrow v' = \pm c\, t^{-(2r+\alpha)}$$

$$= d\, t^{-(2r+\alpha)} \qquad d = \pm c. \qquad (3.46)$$

$$v = d\ln t + f, \quad f \text{ constant}$$

$$y = y_1 v = t^r(d\ln t + f)$$

$$= \underbrace{f t^r}_{y_1} + \underbrace{d t^r \ln t}_{y_2}$$

Ex 3.28

$$t^2y'' + 5ty' + 4y = 0 \qquad t>0$$

$$y = t^r \qquad y' = rt^{r-1} \qquad y'' = r(r-1)t^{r-2}$$

$$r(r-1)t^{r-2} + 5rt^{r-1} + 4t^r = 0$$

$$r(r-1) + 5r + 4 = 0$$

$$r^2 + 4r + 4 = 0 \Rightarrow (r+2)^2 = 0$$

$$r_1 = r_2 = -2$$

general sol^n

$$y = t^{-2}(c_1\ln t + c_2) \qquad t>0$$

Check (i) $y_1 = t^{-2}$ and $y_2 = (\ln t)t^{-2}$ are sol^{ns}

(ii) linearly independent.

§ 3.8.3

$$t^r = e^{r\ln t} \qquad r = \lambda + i\mu$$

$$t^{\lambda+i\mu} = e^{(\lambda+i\mu)\ln t}$$

$$= e^{\lambda\ln t + i\mu\ln t}$$

$$= e^{\lambda\ln t} \cdot e^{i\mu\ln t}$$

$$= e^{\ln t^\lambda} \cdot e^{i\mu\ln t}$$

$$= t^\lambda \cdot e^{i\mu\ln t}$$

Euler identity $e^{i\theta} = \cos\theta + i\sin\theta \qquad \theta = \mu\ln t$

Ex 3.29 $t^2y'' + ty' + y = 0 \quad t > 0$

$y = t^r$

$r(r-1) + r + 1 = 0$

$r^2 + 1 = 0$

$r = \pm i$

Complex conjugate roots $r_1 = \lambda + i\mu \quad \lambda = 0 \quad \mu = 1$

$r_2 = \lambda - i\mu = \bar{r}_1$ complex conjugate.

General sol: $y = c_1 t^0 \cos(\ln t) + c_2 t^0 \sin(\ln t)$ using (3.49)

$= c_1 \cos(\ln t) + c_2 \sin(\ln t), \quad t > 0$

$ay'' + by' + cy = 0 \qquad y = e^{mt}$

$t^2y'' + \alpha t y' + \beta y = 0 \qquad y = t^r$

$\boxed{t = e^s} \qquad \frac{dt}{ds} = e^s \qquad y(t) \quad y(t(s))$

$y' = \frac{dy}{dt} = \frac{dy}{ds} \cdot \frac{ds}{dt}$ chain rule

$\boxed{y' = \frac{dy}{ds} \cdot e^{-s}} \qquad y'' = \ldots$